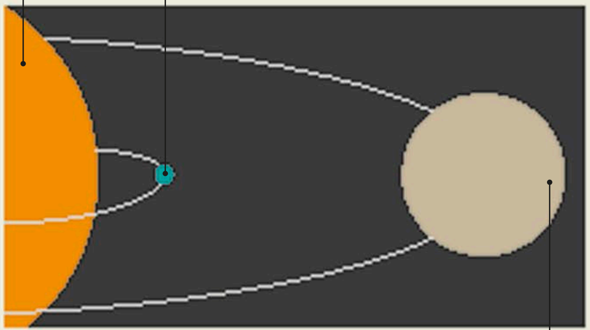
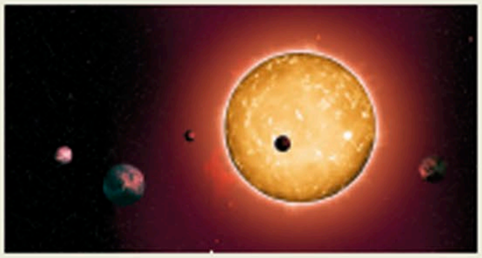


Orbital times

Kepler's discoveries

Johannes Kepler, a German astronomer, set out to prove that Copernicus's theory of a Sun-centered Universe was correct. Rather than orbit the Sun in circles, Kepler found that the planets travel around it in ellipses (oval paths). He also discovered that the farther a planet is from the Sun, the longer it takes to complete its orbit.


Artist's impression of Kepler-444 star system

Planets beyond our solar system

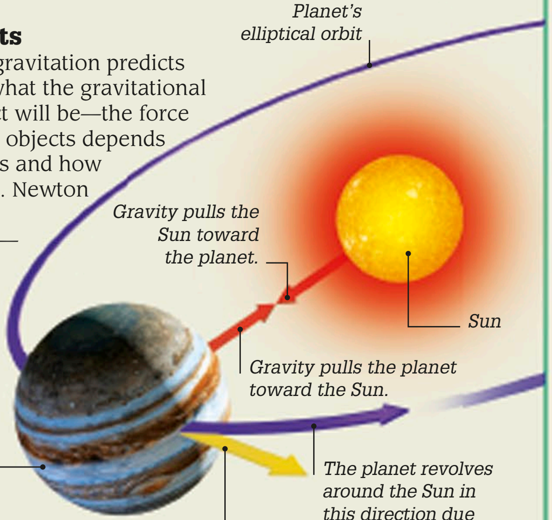
We now know of thousands of planets orbiting parent stars beyond our solar system. They are known as exoplanets. NASA's Kepler space telescope, launched in 2009, is able to detect orbits of exoplanets by measuring how far a star's light dims when a planet passes in front of it. The Kepler-444 star system, discovered by this telescope, contains five planets that orbit their star in less than 10 days.

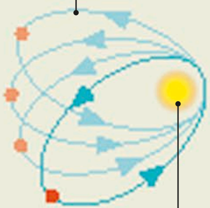
Newton's law of gravity

Astronomers could not explain why the planets follow elliptical orbits until Isaac Newton, the great English physicist, supplied the answer (see p.88). Gravity, which makes an apple fall to the ground on Earth, also keeps the planets in orbit around the Sun. All matter exerts gravity, pulling other matter toward it. The strength of gravity depends on the mass of the object, and weakens with distance.

Elliptical orbits

Newton's law of gravitation predicts mathematically what the gravitational force on an object will be—the force between any two objects depends upon their masses and how far apart they are. Newton worked out the forces on planets—and his equation predicted that orbits would be elliptical, just as Johannes Kepler had observed.





Mercury's wobbly orbit

Over time, the planet Mercury's orbit shifts slightly. Newton's law of gravity could not explain this puzzling fact. In 1915, physicist Albert Einstein solved the mystery in outlining his theory of relativity (see p.173). The wobbly orbit is due to Mercury having to travel through an area of space that is curved due to the Sun's mass.

> **"We revolve around the Sun like any other planet."**
>
> Nicolaus Copernicus

1687

Isaac Newton formulated the universal law of gravitation and explained that it is the force of gravity that holds the planets in elliptical orbits around the Sun.

1781

William Herschel, a British astronomer, discovered the planet Uranus in orbit beyond Saturn. It was the first planet to be discovered since ancient times.

1846

Mathematical calculations correctly predicted the existence of a new planet, later given the name of Neptune, before it was observed by telescope.

2009

NASA launched Kepler, a space observatory, to discover habitable, Earth-sized planets orbiting other stars. By 2016, it had discovered 21 Earth-like planets.